

4. Answer: D

When the object is travelling downwards at velocity 12 m s^{-1} , resultant force

$$mg - R = ma$$

$$(3.0)(9.81) - (0.60)(12) = 3.0a$$

$$a = 7.41 \text{ m s}^{-1}$$

Maximum velocity reached when object travelling at terminal velocity, i.e. air resistance is equal and opposite to its weight.

$$R = mg$$

$$0.60v = mg$$

$$v = \frac{mg}{0.60} = \frac{3.0 \times 9.81}{0.60}$$
$$= 49.1 \text{ m s}^{-2}$$